WORKED SOLUTION

STEP III 2001, Question 11

Mechanics • Moment of inertia

STEP III 2001, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q11

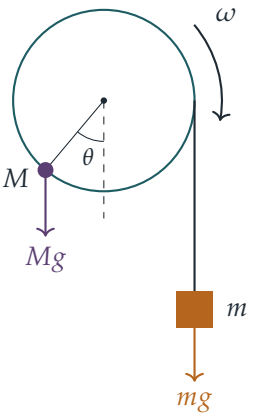
The key idea

Everything is smooth (the axle is free and the string is light), so mechanical energy is conserved, and one energy equation gives ω as a function of θ with no need for tension or forces. The cylinder halts exactly when ω^2 hits zero. So the second part becomes a question about a single function of θ: the hanging mass's loss of potential energy must always beat the particle's gain. Rearranged, that is a comparison between m/M and the largest value of $(1-\cos\theta)/\theta$, which is a straightforward maximisation.

Solution

The angular velocity

We model m and M as particles, the string as light and inextensible and not slipping on the cylinder, and the axle as smooth. Let the cylinder turn through angle θ in the sense that lets m descend, and let $\omega = \dot{\theta}$.



When the cylinder has turned through θ, a length $a\theta$ of string has unwound, so m has

fallen $a\theta$ and moves with speed $a\omega$. The particle M is on the rim, so it moves with speed $a\omega$; it started at depth a below the axis and is now at depth $a\cos\theta$, so it has risen $a(1-\cos\theta)$.

The kinetic energy is

$$\tfrac{1}{2}I\omega^2 + \tfrac{1}{2}M(a\omega)^2 + \tfrac{1}{2}m(a\omega)^2 = \tfrac{1}{2}\big(I + (M+m)a^2\big)\omega^2.$$

No work is done by the smooth axle, and the string, being light and inextensible, transfers energy without loss. So the gain in kinetic energy equals the net loss in potential energy:

$$\tfrac{1}{2}\big(I + (M+m)a^2\big)\omega^2 = mga\theta - Mga(1-\cos\theta),$$

giving

$$\boxed{\omega^2 = \frac{2ga\big(m\theta - M(1-\cos\theta)\big)}{I + (M+m)a^2}} \qquad \text{and} \qquad \omega = \sqrt{\frac{2ga\big(m\theta - M(1-\cos\theta)\big)}{I + (M+m)a^2}}.$$

One modelling point is worth checking: the string must stay taut. Differentiating the energy equation with respect to time gives $\big(I + (M+m)a^2\big)\ddot{\theta} = ga(m - M\sin\theta)$, and the tension is $T = m(g - a\ddot{\theta})$. Now $a\ddot{\theta} < g$ is equivalent to $a^2(m - M\sin\theta) < I + (M+m)a^2$, that is, $-Ma^2\sin\theta < I + Ma^2$, which is always true. So $T > 0$ throughout and the energy equation is valid for all θ.

Never coming to a halt

At the start, $\ddot{\theta} = gam/(I + (M+m)a^2) > 0$, so the cylinder does begin to turn in the direction we chose. Since ω is continuous, it can only change sign by passing through zero, so the cylinder keeps turning for ever provided $\omega^2 > 0$ for every $\theta > 0$. With $k = m/M$, the expression for ω^2 shows that this happens exactly when

$$k\theta > 1 - \cos\theta \quad \text{for all } \theta > 0, \qquad \text{i.e.} \qquad k > h(\theta) = \frac{1-\cos\theta}{\theta} \quad \text{for all } \theta > 0.$$

So we need the largest value of h on $\theta > 0$.

Stationary points. We have

$$h'(\theta) = \frac{\theta\sin\theta - (1-\cos\theta)}{\theta^2}.$$

Using $\sin\theta = 2\sin\tfrac{1}{2}\theta\cos\tfrac{1}{2}\theta$ and $1 - \cos\theta = 2\sin^2\tfrac{1}{2}\theta$,

$$h'(\theta) = \frac{2\sin\tfrac{1}{2}\theta\,\big(\theta\cos\tfrac{1}{2}\theta - \sin\tfrac{1}{2}\theta\big)}{\theta^2}.$$

On $0 < \theta < \pi$ both $\sin\tfrac{1}{2}\theta$ and $\cos\tfrac{1}{2}\theta$ are positive, so $h'(\theta)$ has the sign of $\theta - \tan\tfrac{1}{2}\theta$. Let $u(\theta) = \tan\tfrac{1}{2}\theta - \theta$. Then $u(0) = 0$, $u'(\theta) = \tfrac{1}{2}\sec^2\tfrac{1}{2}\theta - 1$ is negative near 0 and $u''(\theta) = \tfrac{1}{2}\sec^2\tfrac{1}{2}\theta\tan\tfrac{1}{2}\theta > 0$, so u is convex on $[0,\pi)$. It dips below zero and then tends to $+\infty$ as $\theta \to \pi$. A convex function can cross zero at most twice, so u has exactly one zero in $(0,\pi)$, which is α. Hence h increases on $(0,\alpha)$ and decreases on (α,π).

Ask yourself What is the value of h at its maximum, and what about larger θ?

At $\theta = \alpha$ the numerator of h' vanishes, so $1 - \cos\alpha = \alpha\sin\alpha$ and therefore

$$h(\alpha) = \frac{1-\cos\alpha}{\alpha} = \sin\alpha.$$

For $\pi \le \theta \le 2\pi$ we have $\sin\frac{1}{2}\theta \ge 0$ and $\cos\frac{1}{2}\theta \le 0$, so $\theta\cos\frac{1}{2}\theta - \sin\frac{1}{2}\theta \le 0$ and $h' \le 0$: h keeps decreasing up to 2π, so $h < \sin\alpha$ on $(\alpha, 2\pi]$. For $\theta > 2\pi$, $h(\theta) \le 2/\theta < 1/\pi$. Finally, $u(\frac{3}{4}\pi) = \tan\frac{3}{8}\pi - \frac{3}{4}\pi \approx 2.414 - 2.356 > 0$ and $u(\frac{2}{3}\pi) = \sqrt{3} - \frac{2}{3}\pi < 0$, so $\frac{2}{3}\pi < \alpha < \frac{3}{4}\pi$, giving $\sin\alpha > \sin\frac{3}{4}\pi = \frac{1}{\sqrt{2}} > \frac{1}{\pi}$. So $\sin\alpha$ is the largest value of h on $\theta > 0$, attained only at $\theta = \alpha$.

Conclusion. If $m/M > \sin\alpha$, then for every $\theta > 0$

$$m\theta - M(1-\cos\theta) = M\theta\big(k - h(\theta)\big) \ge M\theta(k - \sin\alpha) > 0,$$

so $\omega^2 > 0$ for all $\theta > 0$ and the cylinder rotates without coming to a halt, as required.

(Conversely, if $m/M < \sin\alpha$ then $\omega^2 < 0$ at $\theta = \alpha$, so ω reaches zero before then and the cylinder stops and swings back. In the borderline case $m/M = \sin\alpha$ the system creeps towards the equilibrium position $\theta = \alpha$ without ever quite reaching it.)

Checks

- With $M = 0$ the formula gives $\omega^2 = 2gma\theta/(I + ma^2)$, the standard result for a mass unwinding from a pulley: ω^2 grows linearly, as for constant angular acceleration.
- Solving $\alpha = \tan\frac{1}{2}\alpha$ by bisection gives $\alpha \approx 2.3311$ (about $133.6°$) and $\sin\alpha \approx 0.7246$; a numerical search of $(1-\cos\theta)/\theta$ over $0 < \theta < 100$ gives the same maximum 0.7246.
- Stepping the equation of motion numerically for random I, a, M, m reproduces the formula for ω^2, and with $m/M = 0.735$ the cylinder never stops while with $m/M = 0.715$ it does.

Common slips

- Forgetting the kinetic energy of M or of m: both move with speed $a\omega$, so the effective moment of inertia is $I + (M+m)a^2$.
- Getting the height of M wrong: it rises by $a(1-\cos\theta)$, not $a\sin\theta$.
- Showing only that h has a local maximum at α, without checking that h is smaller for all larger θ.

Beyond STEP

This is one-degree-of-freedom motion in a potential: $\frac{1}{2}J\omega^2 + V(\theta) = 0$ with $V(\theta) = Mga(1-\cos\theta) - mga\theta$, a tilted "washboard" potential. Starting from rest, the motion

runs away for ever exactly when V stays below its starting value for every $\theta > 0$, and the same picture describes a driven pendulum and the Josephson junction in physics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats energy diagrams of this kind.

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